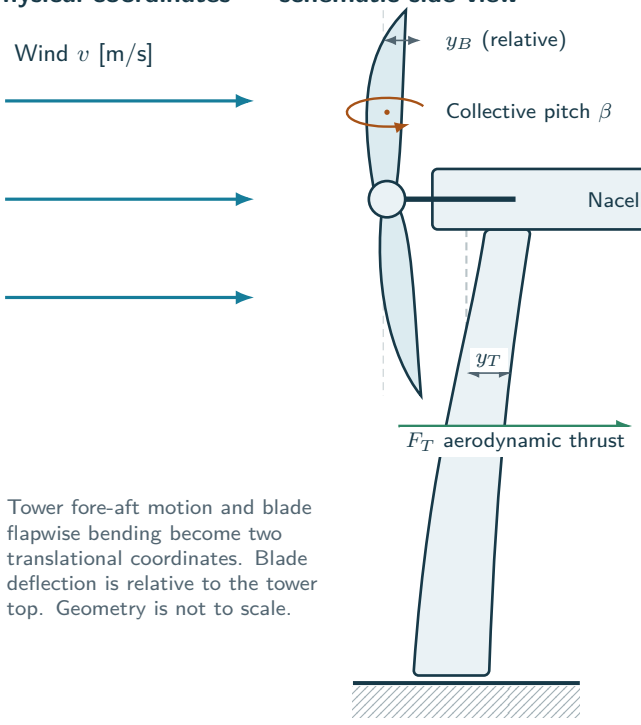
Wind turbine: structure, drivetrain and aerodynamic coupling

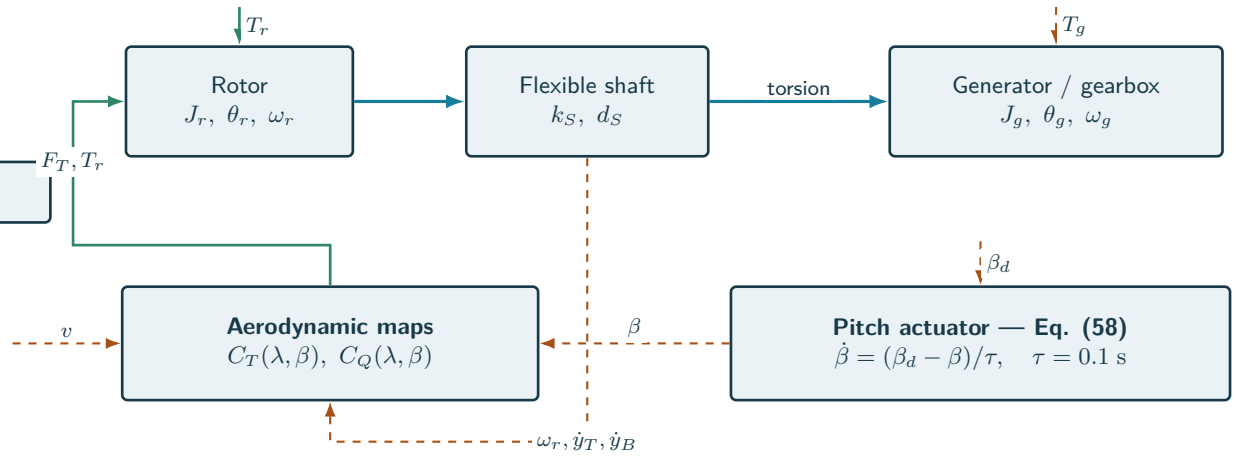
Paper section 3.12: *Wind turbine*

Physical coordinates — schematic side view



Tower fore-aft motion and blade flapwise bending become two translational coordinates. Blade deflection is relative to the tower top. Geometry is not to scale.

Drive-train coordinates and actuator



$$v_{\text{eff}} = v - \dot y_T - \dot y_B, \qquad \lambda = \frac{\omega_r R}{v_{\text{eff}}}, \quad R = 63 \text{ m}$$

$$F_T = \tfrac{1}{2}\rho\pi R^2 C_T v_{\text{eff}}^2, \qquad T_r = \tfrac{1}{2}\rho\pi R^3 C_Q v_{\text{eff}}^2$$

$$M\ddot q + D\dot q + Kq = [F_T, F_T, T_r, -T_g]^{\mathsf T}$$

Generator inertia, angle and torque are referred to the low-speed side. The gearbox ratio is absorbed into these quantities; gearbox dynamics are omitted.

Nine-state reduced model

$q = [y_T, y_B, \theta_r, \theta_g]^{\mathsf T}$.
$x = [q^{\mathsf T}, \dot q^{\mathsf T}, \beta]^{\mathsf T}$.
Eqs. (55)–(58) retain four mechanical degrees of freedom and first-order pitch.

Control-oriented repository variant

The bundled 2022 ROSCO rotor map supplies the aerodynamic coefficients. Local LQR/PDC gains are newly computed; they do not reproduce Table 12.

High-order verification family

The article also names NREL FAST. The repository executes a later OpenFAST 5.0.0 NREL 5 MW case; it is distinct from this nine-state model.

Blue: mechanical propagation or incident wind. Green: aerodynamic force/torque. Solid orange arc: blade-axis pitch. Orange dashed: model signals and commands. The reduced plant has two commands (T_g, β_d), wind disturbance v, and nine states. FAST/OpenFAST is a separate high-order verification model family.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.